

Identification of quantitative trait loci associated with resistance to foliar diseases in sorghum [*Sorghum bicolor* (L.) Moench]

S. Murali Mohan · R. Madhusudhana · K. Mathur ·
D. V. N. Chakravarthi · Sanjay Rathore · R. Nagaraja Reddy ·
K. Satish · G. Srinivas · N. Sarada Mani · N. Seetharama

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Abstract Forage sorghum cultivars grown in India are susceptible to various foliar diseases, of which anthracnose, rust, zonate leaf spot, drechslera leaf blight and target leaf spot cause severe damage. We report here the quantitative trait loci (QTLs) conferring resistance to these foliar diseases. QTL analysis was undertaken using 168 F₇ recombinant inbred lines (RILs) of a cross between a female parental line 296B (resistant) and a germplasm accession IS18551 (susceptible). RILs and parents were evaluated in replicated field trials in two environments. A total of twelve QTLs for five foliar diseases on three sorghum linkage groups (SBI-03, SBI-04 and SBI-06) were detected, accounting for 6.9–44.9% phenotypic variance. The morphological marker Plant color (*Plcor*) was associated with most of the QTL across years and locations. The QTL information generated in this study will aid in the transfer of foliar disease resistance

into elite susceptible sorghum breeding lines through marker-assisted selection.

Keywords *Sorghum bicolor* · Foliar diseases · QTL analysis · Co-localization

Introduction

Sorghum, which originated in tropical and subtropical regions of Africa (Doggett 1988), is an important feed and fodder crop. India contributes about 16% to the world's annual sorghum production (Tonapi et al. 2007). About 60–70% of the total forage produced in India during the rainy season is from sorghum (Grewal 1988). However, several foliar diseases, namely, anthracnose (*Ant*) caused by *Colletotrichum graminicola* (Ces.) Wilson (syn. *C. sublineola* Henn. In Kab. and Bubák), zonate leaf spot (*Zls*) caused by *Gloeocercospora sorghi* (Bain & Edgerton) ex. Deighton, target leaf spot (*Tls*) caused by *Bipolaris sorghicola* (Lefebvre & Sherwin) Alcorn, drechslera leaf blight (*Dlb*) caused by *Drechslera australiensis* (Bugnicourt) Subram. & Jain ex. M.B. Ellis and rust (*Rust*) caused by *Puccinia purpurea* Cooke, are prevalent under warm humid conditions and are highly destructive (Dalmacio 2000; Frederiksen 2000; Mathur and Bunker 2002). These diseases reduce the amount of green leaf area available for photosynthesis and affect the quality of fodder by reducing the protein, zinc and IVDMD (in vitro dry matter digestibility) (Rana et al.

S. Murali Mohan · R. Madhusudhana (✉) ·
D. V. N. Chakravarthi · R. Nagaraja Reddy ·
K. Satish · G. Srinivas · N. Seetharama
Directorate of Sorghum Research, Rajedranagar,
Hyderabad, Andhra Pradesh, India
e-mail: madhu@sorghum.res.in

K. Mathur · S. Rathore
MPUAT, Udaipur, Rajasthan 313 001, India

N. Sarada Mani
Andhra University, Visakhapatnam, Andhra Pradesh,
India

1999). The estimated yield losses due to foliar diseases in Asia, Africa and America range from 32 to 60% (Sharma 1980; Frederiksen 2000). The use of the fungicide to control diseases decreases profit margins of low cash-input forage production systems and increases the risk of residual effects from fungicide on forage. Therefore, the use of host plant resistance is considered to be more practical and reliable for managing foliar diseases. To develop resistant cultivars, plant breeders require a detailed knowledge of the inheritance of resistance. Resistance can be complex, controlled by a single or several genes depending on the source of resistance, plant development stage and the pathotype used (Stefenson et al. 1996; Afanasenko et al. 1999; Cakir et al. 2003).

Quantitative resistance, also referred to as partial resistance, tends to be multigenic and quantitatively inherited (Wastie 1991; Umaerus and Umaerus 1994). Molecular marker technology greatly facilitates the study of complex traits and has made it possible to dissect the polygenes controlling such traits into individual Mendelian factors (Paterson et al. 1988). DNA-based molecular markers delimiting disease resistance loci in sorghum have been reported for head smut (Oh et al. 1994), downy mildew (Gowda et al. 1995; Agrama et al. 2002), leaf blight (Boora et al. 1999), grain mold (Klein et al. 2001), and drechslera leaf blight, target leaf spot, zonate leaf spot (Mohan et al. 2009). Recently, genes for anthracnose resistance in sorghum have been mapped to chromosomes SBI-05 and SBI-08 (Singh et al. 2006; Perumal et al. 2008). The objective of the study reported here was to identify molecular markers tightly linked with quantitative trait loci (QTLs) for resistance to important pathogens causing the above foliar diseases in sorghum.

Materials and methods

Field experiments

The experimental material consisted of a set of 168 F₇ recombinant inbred lines (RILs) derived from the cross between parents, 296B and IS18551. All the RILs (168), parents and susceptible check 'Kekri local' were evaluated for foliar diseases in a randomized complete block design (RCBD) with three replications during the rainy seasons of 2005 and 2007 at the research farm of Rajasthan College of

Agriculture, Maharana Pratap University of Agriculture and Technology, Udaipur (first environment: 24.34°N, 73.4°E, 582.17 m a.m.s.l., mean annual rainfall 550–600 mm), Rajasthan, India, which is a hotspot for of diseases (Anahosur 1992). The second test environment was at the research farm of the Directorate of Sorghum Research (formerly NRCS) located at Rajendranagar, Hyderabad (17.27°N, 78.2°E, 536.0 m a.m.s.l., mean annual rainfall 800–900 mm), Andhra Pradesh, India during the rainy season of 2008. To ensure good disease pressure in the field, the susceptible check 'Kekri local' was grown as infector rows at the borders and also repeated by planting after every 20 test entries. Each entry was planted in a single row plot (4 m long), and the spacing between the rows was 45 cm. The seedlings were thinned at 10 days after emergence (DAE) to a plant-to-plant spacing of 15 cm with a total plant stand of 26 plants per plot. At Udaipur, the whorls of 12- to 25-day-old seedlings of each entry were inoculated with two to three grains of a virulent strain of each pathogen that had been grown for 10 days on autoclaved sorghum grains and air-dried. Artificial inoculations were not carried out for the experiment conducted in the second test environment. Observations of disease severity were recorded at the soft dough stage of grain maturity of the plants, using the standard scale of 1–5 based on the percentage of total leaf area infected, where 1 = no symptoms seen on leaves; 2 = up to 10% of leaf area infected; 3 = 11–25% of leaf area infected; 4 = 26–50% of leaf area infected; 5 = >50% of leaf area infected. *Ant* occurrence was common between 2007 and 2008 (first and second environment, respectively) (Table 4) and that of *Zls* was also common between the two environments (2005 and 2008). *Tls* incidence was limited to 2005 and 2007 (first environment), *Dlb* occurred in 2005 (first environment), whereas *Rust* severity was found only in 2008 (second environment). Although the present mapping population of cross 296B × IS18551 was developed for the primary trait of shoot fly insect pest, which is a major concern of sorghum growers in India and many African countries, the parents are also relatively different for many agronomic traits, including foliar diseases. The aim of our study was to use this available population with parents having relatively different disease responses in order to map the foliar disease QTLs. Efforts are ongoing at our institute for developing mapping populations using contrasting

parents for foliar disease reactions for validation and QTL detection.

The software package Windostat (<http://www.windostat.org>) was used for all statistical analyses of phenotypic data. Broad-sense heritabilities based on the formula $H^2 = \sigma^2_g / \sigma^2_p$ where $\sigma^2_g = (\sigma^2_{Ms_g} - Ms_e) / r$ and $\sigma^2_p = (\sigma^2_g + \sigma^2_e)$. In this formula, σ^2_g = genotypic variance and σ^2_p = phenotypic variance (Singh et al. 1995) were determined from the analysis of variance (ANOVA) across years. Genotypic correlations among traits were estimated using the method of moments (Mode and Robinson 1959).

Marker development using the whole-genome sequence of sorghum

Additional simple sequence repeat (SSR) markers on SBI-06 around the QTLs (Mohan et al. 2009) for foliar diseases were incorporated into the marker dataset based on previous mapping results (Srinivas et al. 2009). The forward and reverse primer sequences of SSRs already mapped that flank or span across foliar disease QTLs were blasted against sorghum genome browser (<http://www.phytozome.net/sorghum>) in order to identify their positions on the sorghum physical map (<http://www.phytozome.net>), and the genomic sequence was downloaded. The parameters set for marker development and PCR amplification conditions were similar to those described in our previous studies (Srinivas et al. 2008) except that in this study we used a minimum of ten repeats as the cut-off for the identification of microsatellite motifs for the development of primer pairs. The amplicons were separated on 6% denaturing polyacrylamide gel electrophoresis and visualized by silver staining (Fritz et al. 1995). The alleles were manually scored in the RILs as A (parent 296B allele), B (parent IS18551 allele) and H (heterozygote-carrying alleles from both parents). The markers were named Fdnhsbm1–Fdnhsbm115 according to the marker nomenclature proposed by De Vicente et al. (2004). Disease-related expressed sequence tags (ESTs) of sorghum were downloaded from the EST public database (<http://www.ncbi.nlm.nih.gov>) and blasted against the sorghum genome (<http://www.phytozome.net/sorghum>) to determine their position on the sorghum physical map.

Linkage map and QTL analyses

The linkage mapping software JOINMAP 3.0 (Van Ooijen and Voorrips 2001) was used to perform chi-square tests of goodness-of-fit to an expected ratio of 1:1 for the segregation of all markers in RILs and to construct the linkage map. The allocation of markers to linkage groups was mostly stable for a wide range of limit of detection (LOD) grouping thresholds (from 4.0 to 7.0). Markers that were attributed to a linkage group at a LOD grouping threshold of 4.0 were only included when it was known that they belong to this group from other published maps. The Kosambi function was used to convert recombination into the genetic distance (centimorgans, cM) (Kosambi 1944). Linkage groups were named according to the nomenclature proposed by Kim et al. (2005).

Three sets of phenotypic data were used (two sets of data from first environment during 2005 and 2007, and the dataset from the second environment during 2008) for QTL analysis of disease resistance. For the analysis, the MapQTL ver. 5 software (Van Ooijen 2005) was used. Putative QTLs were initially identified using interval mapping. The marker closest to each of the QTL peaks was then selected as a co-factor and used in a multiple-QTL model implemented in the MQM mapping [= composite interval mapping (CIM)] procedure of MapQTL. The set of co-factors was adjusted if the most likely position of the QTL differed from that identified in the co-factor selection round, and subsequent rounds of MQM mapping were performed. Markers were removed as a co-factor if their LOD value dropped below the significance threshold. A genome-wide LOD significance threshold ($P < 0.05$) was calculated for each set of phenotypic data using 1000 permutations. The confidence interval for each QTL was set at the 1-LOD support interval. This corresponds approximately to a probability of <0.05 for declaring false positives in the entire genome based on sparse-map model (Lander and Botstein 1989). When separated by a minimum distance of 20 cM (Ungerer et al. 2002) or by a 1-LOD support interval (Lander and Botstein 1989), two peaks on one chromosome were considered as two different QTLs. Otherwise, the higher peak was chosen as the position of the QTL. The position of the QTL was determined as the point of maximum LOD score in the region under consideration. The phenotypic variance explained by a

single QTL was calculated as the square of the partial correlation coefficient (R^2) with the observed variable, adjusted for co-factors.

Results

Phenotypic data

The parental lines exhibited phenotypic differences for field disease assessment, differing in their reaction to foliar diseases (Table 1). Frequency distributions of the severity of foliar diseases (scored at physiological maturity) averaged over three replications of 168 RILs followed continuous variation for all the diseases (Fig. 1). The distribution of RILs was skewed towards the resistant parent 296B for *Ant* and *Dlb*. Transgressive segregation was observed, with few lines showing disease severity values that were less than that of the resistant parent or more than that of the susceptible parent (Table 1; Fig. 1).

Components of variance estimated for the disease severities recorded in the first environment in 2005 and 2007 are presented in Table 2. Across years, the effects of RILs were highly significant ($P < 0.01$). In the ANOVA, the effects of year, replications nested within years, RILs and RILs $\times$ year were highly significant ($P < 0.01$) (Table 2). The RILs $\times$ year interaction was tested against the experimental error. Significant genotype effect to *Tls*, *Zls*, *Dlb* and *Ant* in the field ($P < 0.001$) and significant interaction between RILs

and years ($P < 0.001$) were found (Table 2). Correlations among diseases are presented in Table 3; significant correlations were observed ($P < 0.01$).

Identification of QTLs

QTL for anthracnose resistance (Ant)

The dataset from 2007 (first environment) identified a QTL for *Ant* between Xtxp95 and *Plcor* (Table 4) on SBI-06 explaining a phenotypic variance of 17.7% with a LOD of 6.8. This QTL was not identified from the 2005 dataset. When the average data over 2 years (2005 and 2007) were used, a QTL contributing to *Ant* resistance was identified, which was the same QTL identified using 2007 data alone. This QTL was also identified in the second environment with a LOD value of 12.5; it explained 39.5% of the phenotypic variance. Two additional QTLs from the 2007 dataset alone were identified between *GlumeT*–*mrco* and *Fdnhsbm1*–*Fdnhsbm8*, explaining a phenotypic variance of 8.1 and 6.9%, respectively (Table 4). The QTL identified between *GlumeT* and *mrco* was also identified in the second environment, explaining 8.6% phenotypic variance, with a LOD of 3.2 whereas the QTL between *Fdnhsbm1* and *Fdnhsbm8* was identified only in the first environment. An additional QTL was detected in the second environment on SBI-04 between *Fdnhsbm48* and *Fdnhsbm47*, explaining 10.3% of the phenotypic variance with a LOD of 3.8 (Table 4; Fig. 2).

Table 1 Mean disease rating of parental lines, recombinant inbred lines and heritability estimates (average over years) for sorghum foliar diseases

Trait/sorghum foliar disease ^a	Parental lines		Check Kekri local	RIL population				
	296B	IS18551		Minimum	Maximum	Mean	SD	H^{2b}
<i>Tls</i>	1.2	1.8	3.1	1.0	3.8	1.6	0.3	0.13 ± 0.04
<i>Zls</i>	2.3	2.7	3.7	1.0	4.0	2.4	0.2	0.16 ± 0.05
<i>Dlb</i>	1.0	3.4	3.0	1.0	4.9	1.5	0.5	0.38 ± 0.09
<i>Ant</i>	1.5	2.0	4.4	1.0	3.1	1.5	0.5	0.13 ± 0.04
<i>Rust</i>	1.0	2.0	3.5	1.0	4.5	3.5	0.2	0.36 ± 0.02

RIL recombinant inbred line; SD standard deviation

^a *Ant* anthracnose, caused by *Colletotrichum graminicola* (Ces.) Wilson (syn. *C. sublineola* Henn. In Kab. and Bubák); *Zls* zonate leaf spot caused by *Gloeocercospora sorghi* (Bain & Edgerton) ex. Deighton; *Tls* target leaf spot caused by *Bipolaris sorghicola* (Lefebvre & Sherwin) Alcorn; *Dlb* Drechslera leaf blight caused by *Drechslera australiensis* (Bugnicourt) Subram. & Jain ex. M.B. Ellis; *Rust* rust caused by *Puccinia purpurea* Cooke

^b Broad-sense heritability

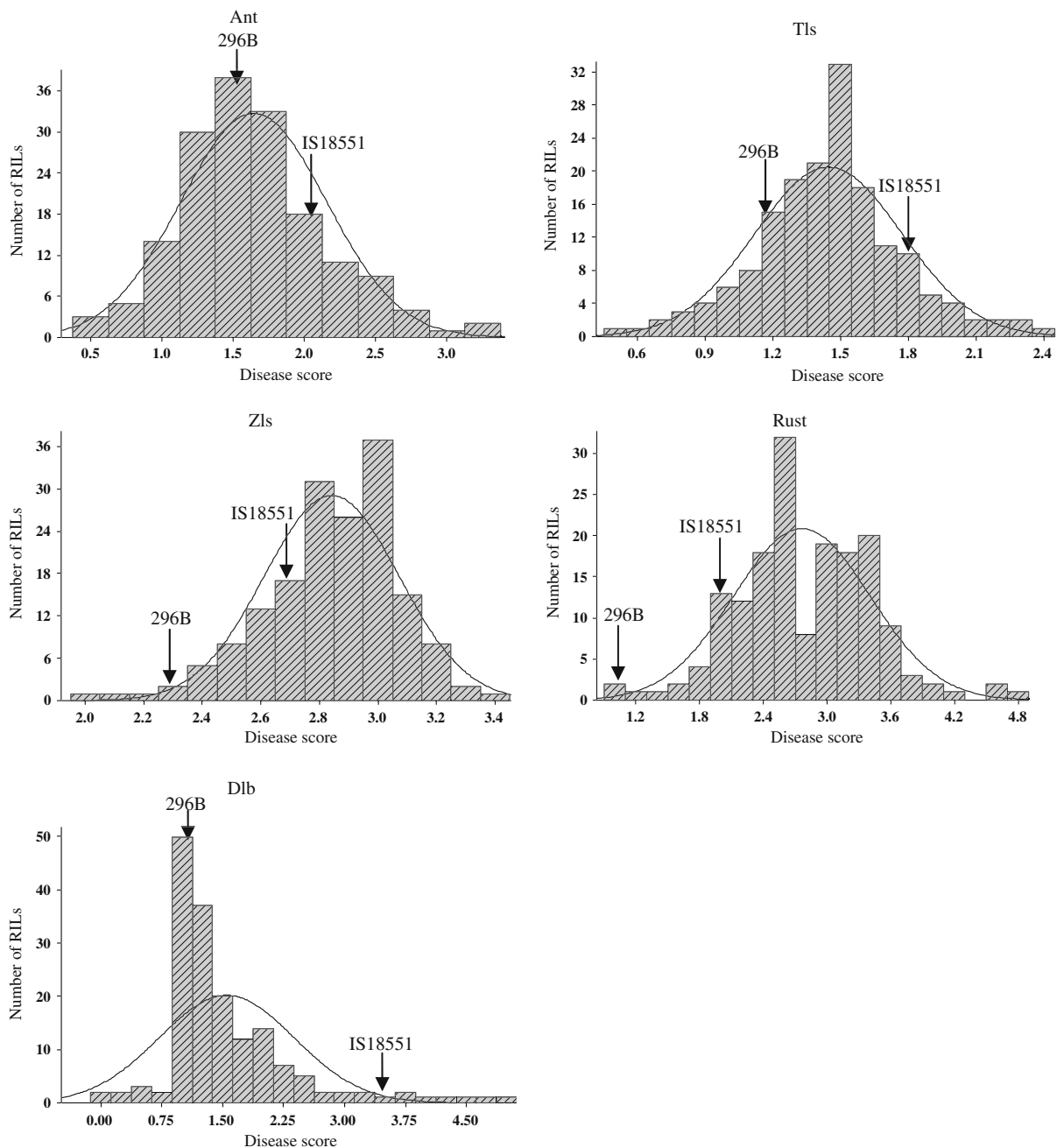


Fig. 1 Frequency distribution of mean disease scores in recombinant inbred lines (RILs) derived from parental lines 296B $\times$ IS18551. For definition of trait/foliar disease, see footnote to Table 1

QTL for zonate leaf spot resistance (Zls)

Of the data for the 2 years in first environment, a QTL for Zls was identified only from the 2005 dataset between Xtxp95 and Plcor on SBI-06. The same QTL was also detected in the second environment.

The QTL from both environments explained 10.2 and 14.2% of the phenotypic variance, respectively (Table 4). The resistance at this locus was derived from the resistant parent 296B. This QTL position corresponds to the QTL position for Ant. Two additional QTLs for Zls resistance were detected in

Table 2 Estimation of variance components for sorghum foliar diseases (2005 and 2007)

Source ^a	<i>Ant</i>		<i>Rust</i>		<i>Tls</i>		<i>Zls</i>		<i>Dlb</i>	
	<i>df</i>	<i>MS</i>	<i>df</i>	<i>MS</i>	<i>df</i>	<i>MS</i>	<i>df</i>	<i>MS</i>	<i>df</i>	<i>MS</i>
Years	1	144.11***	1	30.55*	1	12.97	2	38.66*	1	25.98*
Rep (years)	5	72.05***	5	30.55	8	12.97*	8	38.66	5	12.99*
RILs	167	0.65***	167	0.14	167	3.65*	167	1.09*	167	2.52*
RILs × years	167	72.05***	167	16.32	167	6.48*	167	19.33*	167	12.99*
Error	1725	0.37	1725	0.30	2760	1.77	2760	0.35	1745	0.88

*** Significant at $P < 0.001$ ^a The RILs × year and rep (year) interactions were tested against the experimental error for the combined analysis of variance (ANOVA). The field trial was a randomized complete block design (rep = 3) using 168 progenies (RILs)*df* degrees of freedom, *MS* mean square**Table 3** Genotypic correlations among foliar diseases traits

Trait/sorghum foliar disease	Environment	<i>Zls</i>	<i>Dlb</i>	<i>Ant</i>	<i>Rust</i>
<i>Tls</i>	2005	0.929*	0.449*	–	–
	2007	0.361*	0.181*	0.323*	0.262*
<i>Zls</i>	2005	–	0.555*	–	–
	2007		0.205*	0.214*	0.174*
	2008		–	0.875*	0.905*
<i>Dlb</i>	2005			0.101	–
<i>Ant</i>	2007				0.332*
	2008				0.857*

* Significant at $P \leq 0.01$

–, No disease/not studied

Table 4 Summary of quantitative trait loci for resistance to foliar diseases in sorghum

Trait/sorghum foliar disease	Linkage group	Marker interval	Udaipur (First environment)						Rajendranagar (Second environment)		
			2005			2007			2008		
			LOD	R^2 (%)	Additive effect	LOD	%PV	Additive effect	LOD	%PV	Additive effect
<i>Ant</i>	SBI-06	Xtxp95- <i>Plcor</i>	–	–	–	6.8	17.7	–0.175	12.5	39.5	–0.372
	SBI-06	<i>GlumeT-Mrco</i>	–	–	–	2.9	8.1	–0.083	3.2	8.6	–0.201
	SBI-06	Fdnhsbm1- Fdnhsbm8	–	–	–	3.1	6.9	–0.116	–	–	–
	SBI-04	Fdnhsbm48 -Fdnhsbm47	–	–	–	–	–	–	3.8	10.3	–0.188
<i>Zls</i>	SBI-06	Xtxp95- <i>Plcor</i>	4.3	10.2	–0.151	–	–	–	5.2	14.2	–0.203
	SBI-06	Fdnhsbm107 -Fdnhsbm24	–	–	–	–	–	–	5.4	17.0	–0.281
	SBI-03	Xtxp228- Drenhsbm103	–	–	–	–	–	–	4.1	12.6	–0.186
<i>Tls</i>	SBI-06	Xtxp95- <i>Plcor</i>	19.6	44.9	–0.443	2.3	7.4	–0.129	–	–	–
	SBI-03	Stgnhsbm17- Stgnhsbm9	4.8	9.5	–0.113	–	–	–	–	–	–
<i>Dlb</i>	SBI-06	<i>Plcor</i> -Fdnhsbm93	4.4	11.9	–0.250	–	–	–	–	–	–
<i>Rust</i>	SBI-06	Xtxp95- <i>Plcor</i>	–	–	–	–	–	–	7.5	24.2	–0.339
	SBI-06	Fdnhsbm107 -Fdnhsbm24	–	–	–	–	–	–	5.0	15.3	–0.295

PV phenotypic variance; –, not studied/no disease; LOD logarithmic odds

The nearest flanking marker is given in bold

the second environment. They were flanked by SSR markers Fdnhsbm107–Fdnhsbm24 and Xtxp228–Drenhsbm103, with phenotypic variations of 17.0 and 12.6%, respectively (Table 4).

QTLs for target leaf spot (Tls) and drechslera leaf blight resistance (Dlb)

The dataset from first environment in 2005 was useful to localize a putative QTL for *Tls* on SBI-06 near *Plcor* by a high LOD score of 19.6 explaining 44.9% of the phenotypic variance which was also identified from 2007 data. Another QTL from the same data set was detected between Stgnhsbm17 and Stgnhsbm9 on SBI-03, explaining 9.5% of phenotypic variance supported by a LOD of 4.8. The QTL for *Dlb* was identified only in the first environment (2005) in the same genomic region between Xtxp95 and *Plcor*.

QTLs for Rust resistance

MapQTL detected two QTLs for *Rust* using the dataset from the second environment on SBI-06. The QTL identified between Xtxp95 and *Plcor* with a LOD of 7.5 explained 24.2% of the phenotypic variation. This QTL was not detected in the first environment. Another QTL for *Rust* was detected between Fdnhsbm107 and Fdnhsbm24 with a LOD of 5.0. This QTL explained 15.3% of the phenotypic variation. The two QTLs together explained 39.5% of phenotypic variation in the 2008 experiment.

BLAST results of disease-related ESTs

A total of 19 ESTs showed sequence similarity to the genomic region on sorghum chromosome SBI-06, where a cluster of QTLs for various foliar diseases had been identified in our earlier study (Mohan et al. 2009). The putative function of these ESTs was determined using the BLASTX function, and proteins related to disease resistance were identified (Table 5).

Discussion

In this study, we conducted QTL mapping for resistance to five different biotrophic pathogens. Parental lines exhibited phenotypic differences in disease resistance to all the diseases and there were

correlations among the diseases. The transgressive segregation observed for the traits supports the observation that QTLs for these traits are attributable to both the parents. The transgressive segregation observed in RILs beyond the parental values indicated that both parents carried minor genes or alleles for resistance that differed from each other (Koch and Parlevliet 1991; Young 1996). Thus, even susceptible genotypes may harbor some favorable alleles responsible for resistance and can be helpful for gene pyramiding. There was no disease development for the experiment conducted in the year 2006 (data not provided) due to incessant rainfall for almost one fortnight, due to which inoculations could not be done that resulted in low inoculum, and also the low temperature in this year. Being a quantitative trait, disease resistance is sensitive to genotype x environmental interactions (Geiger and Heun 1989) and consistent heavy disease pressure is required to accurately determine the magnitude of the genetic factor that contributes to the resistance. The expression of certain resistance genes such as *L1*, *L3*, *L7*, *L8*, *L10* and *L11* in flax to flax rust (Islam and Mayo 1990), *Sr6* in wheat to stem rust, *Lr16* and *Lr17* in wheat to leaf rust and many other genes (Browder 1985) are reported to depend on the temperature, to which they are exposed.

Analysis of variance revealed that there were significant differences among RILs ($P < 0.001$) as well as among replicates ($P < 0.001$), suggesting that the RILs differ for resistance to foliar diseases, and that replication of tested RILs was effective in reducing experimental errors. High correlations among disease scores of *Tls*, *Zls*, *Rust* and *Ant* (Table 3) indicated that the resistance to these fungal pathogens could be under common genetic control as reported in other crops like wheat (Lillemo et al. 2007). Therefore, simultaneous selection for potentially durable resistance to four of the important foliar pathogens in sorghum, that are also known to occur in mix infections in nature (Sharma 1980), may be possible by field testing through indirect selection.

QTL analysis

We conducted QTL mapping for resistance to five different biotrophic pathogens in two test environments. In the first environment, artificial inoculum was used for the 2 years (2005 and 2007), and in the second environment (2008), screening was performed

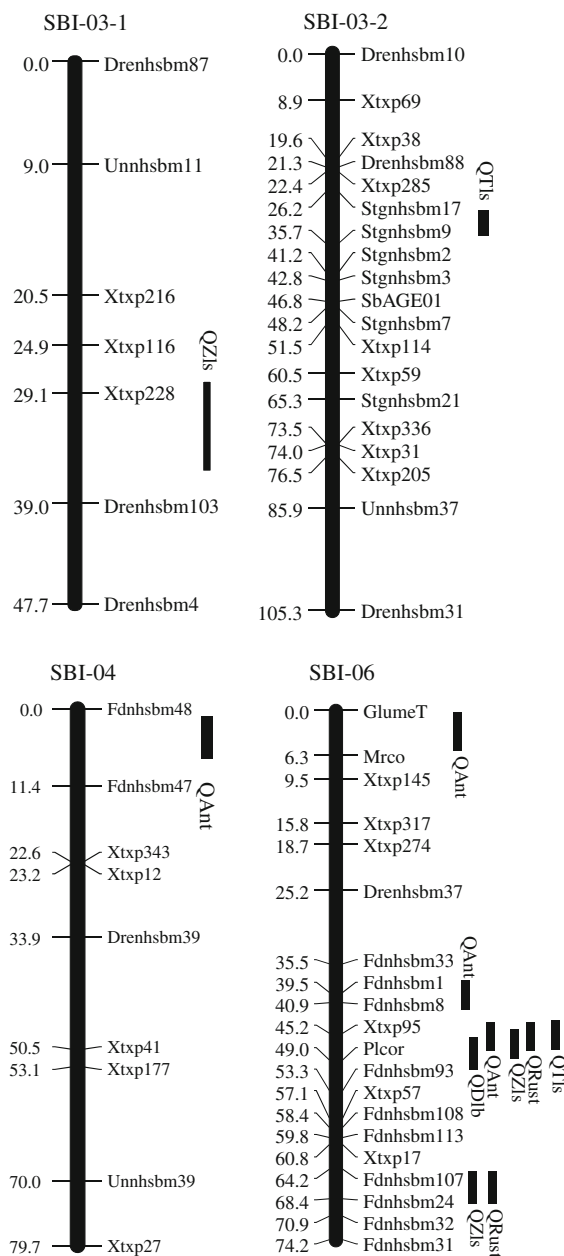


Fig. 2 Linkage groups of sorghum showing quantitative trait loci (QTLs) identified for the foliar diseases

under the conditions of a natural pathogen population. The identification of QTLs for at least three foliar diseases (*Ant*, *Zls* and *Rust*) near the *Plcor* locus with natural inoculum in 2008 confirmed the consistent expression of QTLs under both artificial and natural conditions even though the pathogen population was a mixture of different pathotypes.

QTLs with a LOD <3.0 were reported when they were detected consistently in the same position between environments or years (Table 4). The QTL for target leaf spot resistance detected in the 2005 experiment was confirmed in 2007 by detecting a consistent QTL at the same *Plcor* locus on SBI-06, and this QTL is likely to be stable because it was detected in different years. The other QTL for target leaf spot resistance on SBI-03 was found to be dependent on disease pressure, resulting in low reproducibility. QTLs for disease resistance commonly depend on experimental conditions, plant age (Richter et al. 1998; Chang and Hwang 2003) and race composition of the inoculum (Zhu et al. 2003; Portyanko et al. 2005). This interdependence was supported by the results obtained in our study: QTLs detected in 2005 for *Tls* were not detected in 2006, even though inoculum preserved from the previous year was used. Among the QTLs reported for *Zls*, only one QTL was stable across environments, an observation supported by Mingeot et al. (2002) in wheat. As demonstrated in our study, the important genetic factors behind the *Zls* resistance in the population were located at the *Plcor* locus on chromosome SBI-06. Previous studies (Klein et al. 2001) have detected important QTLs for *Zls* resistance in chromosomal segments corresponding to these loci. The QTL on SBI-06 corresponds to a similar QTL detected in the Sureño and RTx430 mapping population (Klein et al. 2001).

QTLs for rust resistance on SBI-06 detected in our study have not been detected in earlier studies (Tao et al. 1998) and, interestingly, they were co-localized with the cluster of QTLs on SBI-06 for other foliar diseases. The position of this cluster in our study is different from that reported by Tao et al. (1998). This could be due to genetic differences in parental lines, size of the mapping populations and the method of QTL detection and is also probably due to the differences in pathotypes. Pathogenic variability in populations of leaf blight and anthracnose pathogens has been reported at these two test sites (Bunker and Mathur 2006; Thakur et al. 2007), although such information about other pathogens is lacking. Increasing the marker density through the addition of SSR markers has allowed a more accurate estimation of the location of QTLs. QTLs associated with resistance to all five diseases on SBI-06 were flanked by markers within a genetic distance of 4 cM.

Table 5 Putative function of sorghum ESTs detected on SBI-06 near QTL cluster

Sequence no.	EST gene bank accession no.	EST position on the sorghum physical map	BLASTX putative function	E value
1	BM324257	Chromosome_6: 51.6 Mbp	Beta-glucosidase (<i>Oryza sativa</i>)	1.00E–79
2	BE364917	Chromosome_6: 54.1 Mbp	Putative reverse transcriptase (<i>Sorghum bicolor</i>)	3E–15
3	BM327524	Chromosome_6: 54.8 Mbp	Hypothetical protein (<i>Oryza sativa</i>)	1.00E–37
4	BE364241	Chromosome_6: 55.1 Mbp	Ubiquitin-dependent protein (<i>Oryza sativa</i>)	2E–38
5	BM331243	Chromosome_6: 55.1 Mbp	Zn finger containing protein(<i>Oryza sativa</i>)	3E–06
6	BE601016	Chromosome_6: 55.2 Mbp	Beta-carotene hydroxylase (<i>Glycine max</i>)	5.00E–47
7	BE365353	Chromosome_6: 55.5 Mbp	Calcium-dependent protein kinase (<i>Zea mays</i>)	2.00E–58
8	CB928672	Chromosome_6: 55.7 Mbp	Putative purine permease (<i>Oryza sativa</i>)	3.00E–29
9	CD422894	Chromosome_6: 55.8 Mbp	MRP-like ABC transporter (<i>Oryza sativa</i>)	3.00E–108
10	BE364261	Chromosome_6: 56.2 Mbp	Scramblase-like (<i>Oryza sativa</i>)	9.00E–47
11	CD423721	Chromosome_6: 56.6 Mbp	Ubiquitin thiolesterase (<i>Arabidopsis thaliana</i>)	9.00E–45
12	BE595483	Chromosome_6: 56.8 Mbp	Putative protein phosphatase type 2C (<i>Oryza sativa</i>)	1.00E–30
13	BM326537	Chromosome_6: 56.9 Mbp	Hypothetical protein (<i>Oryza sativa</i>)	3.00E–74
14	BM318434	Chromosome_6: 57.0 Mbp	Ent-kaurene synthase 1 (<i>Oryza sativa</i>)	2.00E–66
15	CD424405	Chromosome_6: 57.4 Mbp	Putative LRR transmembrane protein kinase (<i>Arabidopsis thaliana</i>)	4.00E–06
16	BM325100	Chromosome_6: 57.6 Mbp	SIT4 phosphatase-associated family protein (<i>Arabidopsis thaliana</i>)	3.00E–67
17	BM318689	Chromosome_6: 57.7 Mbp	60S ribosomal protein l27 (<i>Oryza sativa</i>)	3.00E–25
18	BE364807	Chromosome_6: 57.7 Mbp	Unknown protein	Not significant
19	BM327086	Chromosome_6: 60.0 Mbp	GTPase (<i>Oryza sativa</i>)	6.00E–33

EST expressed sequence tag; Mbp million base pairs

Co-localization of QTLs

Comparison of the common markers on SBI-06 with the consensus map of rice chromosome 4 (<http://www.gramene.org/cmap>) and the literature revealed that some of the detected QTLs co-localized with known resistance genes and that markers linked with QTLs on other linkage groups in this study may be new resistance loci. The co-localization of QTLs for two diseases, *Ant* and *Rust*, was also observed near *Plcor* locus in addition to the co-location of *Tls*, *Zls* and *Dlb*, as reported in our earlier report (Mohan et al. 2009). Co-localization of QTLs for various diseases (grain mould, anthracnose, ZLS and bacterial leaf stripe) was also reported by Klein et al. (2001). Plant color mapped to SBI-06, with tan color (296B allele) linked in coupling phase with QTLs influencing the severity of foliar diseases. A relationship between tan plants and resistance to foliar and panicle diseases has been documented in sorghum (Rana et al. 1976; Torres-Montalvo et al.

1992). It has been a practice in India under national sorghum breeding programs to select for plant color (tan plant type) as a means to select for resistance to most of the foliar diseases in general. Nicholson et al. (1987) reported the accumulation of phytoalexins (flavonoids) in sorghum in response to pathogen infection. Genes conferring resistance to different pathogens as well as to different races of the same pathogen have been reported to be clustered in chromosomal regions in many plant species (Vear et al. 1997; Sharma et al. 2004).

In addition to the QTLs related to disease resistance identified in our study on SBI-06, this region also contains resistance genes such as the *Rp1* (AAM03018) of maize for rust resistance, the *Pc* gene (EU583216) for resistance to sorghum milo disease (root and crown rot) caused by *Periconia circinata* (Mangin) Sacc., a soil-borne saprophytic fungus (Leukel 1948), the *Pi37* (ABI94578) gene conferring resistance to rice blast fungus *Pyricularia grisea* Sacc. and the sorghum *Rp1* (AAQ74890) gene, a homolog of

the maize rust disease resistance locus *Rp1* that confers resistance to *Puccinia sorghi*. The clustering of resistance loci and synteny between different members of the Gramineae family suggest that common loci may be involved in conditioning resistance and that the genes underlying these QTLs may have evolved from a common ancestral gene during evolution. The clustering of functionally related genes has been of great interest in many species in the recent past, and the cluster phenomenon has been explained by the adaptive significance of the functional gene network. It is reasoned that the mechanism for QTL clustering may be due to tight linkage and/or a pleiotrophic effect of the gene/s involved in the expression of genetically related traits. The coincidental mapping of several QTLs is also possibly due to an effect of the small sample size on mapping resolution and scant recombination within blocks of QTL genes (Peng et al. 2003). QTL studies carried out in several cereal crops (Xiong et al. 1999; Bres-Patry et al. 2001; Peng et al. 2003; Cai and Morishima 2002; Upadhyayula et al. 2006; Srinivas et al. 2009; Satish et al. 2009) have demonstrated the clustering of QTLs for the related traits. Chromosomal regions where QTL clusters were found coincided with the regions harboring genes or gene blocks where the frequency of cultivar-derived alleles in RILs was higher than expected. This distortion may be partly due to unconscious selection favoring a cultivated plant type during the establishment of RILs. The chromosomal regions showing apparent over-representation of cultivar alleles in RILs were associated with QTL clusters. It can be inferred from these results that a predominance of cultivar-derived alleles may largely reflect the selective advantage of that cultivar under the cultivation environment in which the RI population was developed. The commonalities of the resistance genes in sorghum with those in maize and rice may be useful in finding and characterizing resistance genes. Since the genetic distance between the QTL and the flanking markers is very small (<4 cM), markers and QTLs have limited probability of recombination. This could be a major factor in utilizing these flanking markers to achieve success in marker-assisted selection for foliar disease resistance. Additional research will be necessary to determine the specificity and agronomic utility of these resistance QTLs as well as their relationships to other plant disease resistance genes. Genetic dissection of chromosomal regions

putatively associated with multiple disease resistance will help breeders gain an understanding of the true linkage and pleiotropy.

ESTs related to compatible interaction

Disease-related ESTs (BM324257, BM327524, BM327086, BE325100 and BM326537) (Table 5) found near the QTL cluster region on SBI-06 are related to the compatible interaction by the compatible strain of *Colletotrichum graminicola* (details as given in EST library <http://www.ncbi.nlm.nih>). This reaction may be similar to the expression of the *Xa1* gene (Yoshimura et al. 1983) in rice that confers resistance to blight disease upon wounding and infection by both compatible and incompatible strains of *Xanthomonas oryzae* pathovar *oryzae* (*Xoo*). ESTs related to salicylic acid induction (CD422894, CD423721) were considered to be the systemically transported signal for systemic acquired resistance (Ryals et al. 1996). Leucine-rich repeat (LRR) transmembrane proteins (CD424405) have been identified to be involved in the recognition of the pathogen (Ellis et al. 2000). GTPase (BM327086) is involved in the regulation of the production of reactive oxygen intermediates, such as superoxide and hydrogen peroxide (H_2O_2), which play an important role in signal transduction leading to activation of defense responses (Bolwell 1999). ABC transporters (CD422894) are involved in the detoxification of pathogen toxins (Stein et al. 2006). Protein phosphatases (BM325100) are involved in disease resistance-related plant responses in the pathogen recognition. Calcium-dependent protein kinases play active role in cell-surface recognition of a pathogen ligand and subsequent activation of intracellular kinase, leading to a defense response. Beta-glucosidases (BM324257) are related to pathogen defense (Brader et al. 2007), and Zn finger-containing protein (BM331243) is thought to be involved in protein–protein interactions (Szabo et al. 1996), which play an important role in signal transduction (Pawson and Scott 1997) and plant stress/defense responses.

Partial-resistance QTLs frequently co-localize with known R-gene loci, and it is assumed that many QTLs correspond to weak or defeated R-genes—in most cases, the canonical nucleotide binding site–LRR domains (Gebhardt and Valkonen 2001). This finding supports our detection of genes coding for

LRR-containing proteins close to the QTL cluster identified on SBI-06.

In conclusion, consistent QTLs (two for *Ant*, one each for *Zls* and *Tls*) on SBI-06 were detected across environments or years. These consistent QTLs on SBI-06 flanked by Xtxp95 and Fdnhsbm93 near plant color could be the ideal target regions for the marker-assisted transfer of disease resistance QTLs into susceptible high-yielding lines. The information obtained from preliminary synteny analysis in this study will facilitate the identification of candidate genes, leading to the map-based cloning of resistance genes for foliar diseases.

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